

Práctica: Comparación de perfiles K-mer

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Ejercicio 1

Se tienen las siguientes secuencias:

- S1 = AGACTG
- S2 = GGATCCAG
- S3 = ATCAA
- S4 = AAATGC

Calcular la similitud de coseno utilizando 2 k-mers. Indique que secuencias son los vecinos más próximos. Justifique su respuesta.

* Extraer los K-mers (K=2)

S1 = AG, EA, AC, CT, TG
 S2 = GG, GA, AT, TC, CC, CA, AG
 S3 = AT, TC, CA, AA
 S4 = AA, AA, AT, TG, GC

* Comparación de los K-mers

2-mer	f ₁	f ₂	f ₃	f ₄	P ₁	P ₂	P ₃	P ₄
AG	1	1	0	0	2/17	2/19	1/16	1/17
BA	1	1	0	0	2/17	2/19	1/16	1/17
AC	1	0	0	0	2/17	1/19	1/16	1/17
CT	1	0	0	0	2/17	1/19	1/16	1/17
TG	1	0	0	1	2/17	1/19	1/16	2/17
GG	0	1	0	0	1/17	2/19	1/16	1/17
AT	0	1	1	1	1/17	2/19	2/16	2/17
TC	0	1	1	0	1/17	2/19	2/16	1/17
CC	0	1	0	0	1/17	2/19	1/16	1/17
CA	0	1	1	0	1/17	2/19	2/16	1/17
AA	0	0	1	2	1/17	1/19	2/16	3/17
GC	0	0	0	1	1/17	1/19	1/16	2/17
	5	7	4	5	1	1	1	1

* Calculando producto escalar

$$|P_1| = \sqrt{\frac{2^2+2^2+2^2+2^2+2^2+1^2+1^2+1^2+1^2+1^2+1^2+1^2}{17^2}} = \frac{\sqrt{27}}{17} = 0.3056$$

$$|P_2| = \sqrt{\frac{2^2+2^2+1^2+1^2+1^2+2^2+2^2+2^2+2^2+2^2+1^2+1^2}{19^2}} = \frac{\sqrt{33}}{19} = 0.3023$$

$$|P_3| = \sqrt{\frac{1^2+1^2+1^2+1^2+1^2+1^2+2^2+2^2+1^2+2^2+2^2+1^2}{16^2}} = \frac{\sqrt{24}}{16} = 0.3061$$

$$|P_4| = \sqrt{\frac{1^2+1^2+1^2+1^2+2^2+1^2+2^2+1^2+1^2+1^2+3^2+2^2}{17^2}} = \frac{\sqrt{29}}{17} = 0.3167$$

* Comparación

$$\cos(P_1, P_2) = \frac{1}{17 \cdot 19} (2 \cdot 2 + 2 \cdot 2 + 2 \cdot 1 + \dots + 1 \cdot 2 + 1 \cdot 1 + 1 \cdot 1) = \frac{\sqrt{27} \times \sqrt{33}}{17 \cdot 19} = 0.871$$

$$\cos(P_1, P_3) = \frac{1}{17 \cdot 16} (2 \cdot 2 + 2 \cdot 1 + 2 \cdot 1 + \dots + 1 \cdot 1 + 1 \cdot 2 + 2 \cdot 1) = \frac{\sqrt{27} \cdot \sqrt{24}}{17 \cdot 16} = 0.825$$

$$\cos(P_1, P_4) = \frac{1}{17 \cdot 17} (2 \cdot 1 + 2 \cdot 1 + \dots + 1 \cdot 1 + 2 \cdot 2) = \frac{\sqrt{27} \cdot \sqrt{29}}{17^2} = 0.822$$

$$\cos(P_2, P_3) = \frac{1}{19 \cdot 16} (2 \cdot 2 + 2 \cdot 1 + \dots + 2 \cdot 2 + 1 \cdot 1) = \frac{\sqrt{33} \times \sqrt{24}}{19 \cdot 16} = \boxed{0.924} \text{ Mayor similitud}$$

$$\cos(P_2, P_4) = \frac{1}{19 \cdot 17} (2 \cdot 1 + 2 \cdot 1 + \dots + 2 \cdot 1 + 1 \cdot 2) = \frac{\sqrt{33} \times \sqrt{19}}{19 \cdot 17} = 0.808$$

$$\cos(P_3, P_4) = \frac{1}{16 \cdot 17} (2 \cdot 3 + 1 \cdot 1 + \dots + 2 \cdot 1 + 1 \cdot 2) = \frac{\sqrt{24} \times \sqrt{29}}{16 \cdot 17} = 0.910$$